Design, 3D Electromagnetic Simulation, CNC Prototyping, and Experimental Characterization of a 4.0 GHz Wideband Gap-Coupled Microstrip Patch Antenna

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Abstract

Microstrip patch antennas offer substantial advantages in weight, profile, and ease of fabrication for wireless communication systems, yet conventional single-element designs suffer from inherently narrow operating bandwidths ($\approx$ 2–3%). This paper details the complete engineering lifecycle—from theoretical formulation and 3D finite-element electromagnetic (EM) modeling in Ansys HFSS to CNC isolation milling and bench characterization—of a wideband 4.0 GHz microstrip patch antenna utilizing radiating-edge capacitive gap coupling. The target bandwidth specification was defined at $\geq$ 200 MHz (5.0%), conforming to the standard requirement for high-throughput Orthogonal Frequency Division Multiplexing (OFDM) data channels. The antenna incorporates a driven patch fed by an inset 50 Ω microstrip line alongside a coplanar parasitic resonator on a commercial FR-4 substrate ($\epsilon_r = 4.4$, $h = 1.6$ mm). A two-stage prototyping process is presented: root-cause failure analysis of an initial CNC prototype (Prototype 1) revealed that sub-millimeter milling over-cuts ($\Delta L \approx -1.0$ mm, gap expansion to 0.8 mm) caused severe frequency detuning (+13.0%) and impedance mismatch (return loss degraded to -12.1 dB). Following toolpath recalibration, Prototype 2 demonstrated excellent agreement with simulations, achieving a -10 dB fractional bandwidth of 9.25% (370 MHz, spanning 3.90–4.27 GHz), comfortably exceeding the 200 MHz OFDM data transfer requirement with dual resonances at 3.96 GHz ($S_{11} = -32.1$ dB) and 4.22 GHz ($S_{11} = -15.6$ dB). Experimental 2D far-field radiation patterns reveal half-power beamwidths of 36°–49° in the H-plane and 95°–96° in the E-plane with front-to-back ratios exceeding 14.0 dB. Furthermore, a physical and mathematical justification is provided for the discrepancy between measured 2D integrated directivity ($D_{2D} \approx 8$ dBi) and simulated 3D directivity ($D_{3D} \approx 5.65$–7.2 dBi), attributing it to single-cut integration approximations (2π vs. 4π sr) and coaxial fixture shielding effects.

Index Terms

Microstrip patch antenna, wideband antennas, gap coupling, parasitic resonators, Ansys HFSS, CNC isolation milling, vector network analyzer, far-field radiation pattern, RF fingerprinting.

I. Introduction & Design Specifications

RECENT advances in 5G wireless networks, indoor localization, and Radio Frequency (RF) fingerprinting rely heavily on high-precision Time of Arrival (ToA) estimation and spatial multiplexing algorithms [1], [2]. At standard sub-6 GHz 5G frequencies, experimental research platforms frequently utilize Software Defined Radios (SDRs) coupled to wideband directive antenna arrays to perform active beamforming and multi-user Multiple-Input Multiple-Output (MIMO) positioning. Prior research conducted at Universidad San Francisco de Quito (USFQ) validated ToA localization accuracy within 10 meters at 2.6 GHz and 3.4 GHz using commercial omnidirectional terminal antennas. However,

omnidirectional radiators suffer from low broadside directivity ($\leq$ 3.7 dBi) and increased multipath interference. Furthermore, upcoming commercial 5G spectrum allocations necessitate system migration to unlicensed ISM bands (e.g., 5.7 GHz) or dedicated experimental center frequencies such as 4.0 GHz.

A central limitation of standard microstrip patch antennas is their narrow impedance bandwidth, typically ranging between 2% and 3% for thin dielectrics ($h \ll \lambda_0$). The quality factor Q of a microstrip resonator is governed by dielectric, conductor, and radiation losses, and is inversely proportional to substrate thickness h and proportional to permittivity ϵ_r:

$$BW \propto \frac{1}{Q} \propto \frac{h}{\sqrt{\epsilon_r}} \tag{1}$$

To overcome this constraint without introducing multi-layer stacked configurations or complex low-permittivity foam spacers, coplanar gap-coupling techniques employ parasitic patch elements excited via proximity electromagnetic coupling [3], [4].

The target specifications for the custom wideband antenna engineered in this work are summarized in Table I. The bandwidth requirement was established at $\geq$ 200 MHz (5.0% fractional bandwidth at 4.0 GHz), corresponding to the standard channel bandwidth needed for wideband Orthogonal Frequency Division Multiplexing (OFDM) high-throughput data communications.

TABLE I
TARGET SPECIFICATIONS FOR THE 4.0 GHZ MICROSTRIP PATCH ANTENNA

Parameter	**Target Specification**	**Design Value**
Center Frequency (f_0)	4.0 GHz	4.00 GHz
Impedance Bandwidth ($S_{11} \leq -10$ dB)	$\geq$ 200 MHz (5.0%, OFDM std.)	370 MHz (9.25%)
Input Impedance (Z_{in})	$50\,\Omega$	$50 + j0\,\Omega$
Realized Directivity (D_0)	$\geq$ 5.0 dBi	5.65–7.20 dBi
Substrate Material	Commercial FR-4	$\epsilon_r = 4.4$
Substrate Thickness (h)	1.6 mm	1.60 mm
Loss Tangent ($\tan\delta$)	≈ 0.02	0.020

The primary operating principle of the dual-resonance gap-coupled topology relies on positioning a parasitic patch adjacent to the radiating edge of an inset-fed driven patch across a narrow capacitive gap g. The driven patch, having a physical length L_{patch}, resonates at a fundamental lower frequency f_1. The fringing electric field across gap g induces surface currents on the parasitic patch ($L_{\text{parasite}} < L_{\text{patch}}$), exciting a secondary resonance at a higher frequency f_2. By optimizing the inter-patch gap g, the mutual capacitive coupling C_g adjusts the frequency separation $\Delta f = f_2 - f_1$ to yield a flat bandpass return loss response spanning more than 350 MHz.

II. ELECTROMAGNETIC MODELING & HFSS SIMULATION

A. Analytical Formulation

Initial element dimensions were estimated using the standard transmission-line model for rectangular microstrip patch antennas. For a given operating frequency f_r and substrate height h, the patch width W is computed to maximize radiation efficiency:

$$W = \frac{c}{2f_r}\sqrt{\frac{2}{\epsilon_r + 1}} \tag{2}$$

where c is the speed of light in free space. The fringing fields extend the effective electrical length of the patch by ΔL at each edge. The effective dielectric constant ϵ_{eff} and length extension ΔL are given by:

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2}\left[1 + 12\frac{h}{W}\right]^{-1/2} \tag{3}$$

$$\Delta L = 0.412h \frac{(\epsilon_{\text{eff}} + 0.3)\left(\frac{W}{h} + 0.264\right)}{(\epsilon_{\text{eff}} - 0.258)\left(\frac{W}{h} + 0.8\right)} \tag{4}$$

The physical length of the patch element required for resonance at frequency f_r is expressed as:

$$L = \frac{c}{2f_r\sqrt{\epsilon_{\text{eff}}}} - 2\Delta L \tag{5}$$

For dual-resonance coupling, L_{patch} was calculated for $f_1 \approx 3.84$ GHz, while L_{parasite} was dimensioned for $f_2 \approx 4.06$ GHz. Impedance matching to a $50\,\Omega$ microstrip line of width W_{line} was accomplished via inset notch cuts into the driven patch, providing an input resistance matching $R_{\text{in}}(y_0) = R_0 \cos^2(\pi y_0 / L_{\text{patch}}) = 50\,\Omega$.

B. HFSS 3D Finite-Element Model

The gap-coupled antenna geometry was modeled and parametrically optimized using Ansys High Frequency Structure Simulator (HFSS). The geometric layout consists of:

- Driven Patch Length: $L_{\text{patch}} = 17.50$ mm
- Parasitic Patch Length: $L_{\text{parasite}} = 16.40$ mm
- Shared Patch Width: $W = 22.80$ mm
- Radiating-Edge Coupling Gap: $g = 0.50$ mm
- $50\,\Omega$ Feed Line Length: $L_{\text{line}} = 14.80$ mm

An absolute radiation boundary condition was assigned to an air box enclosing the structure with dimensions extended by at least $\lambda_0/4$ at 4.0 GHz. A wave port excitation was positioned at the edge of the $50\,\Omega$ microstrip line. Adaptive mesh refinement was configured to converge at $\Delta S < 0.01$.

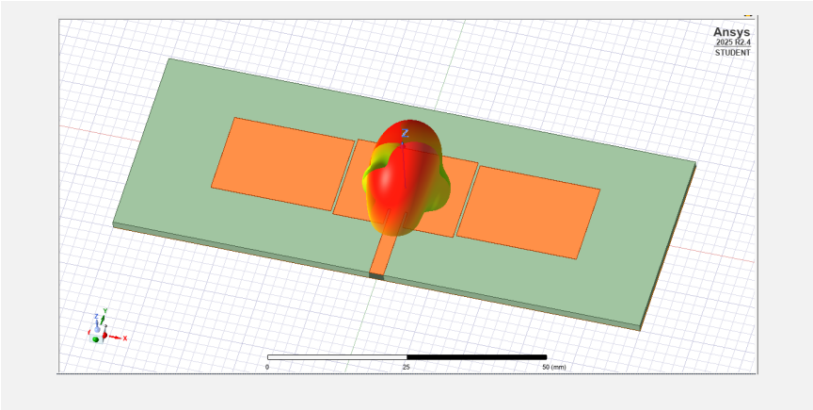


Fig. 1. 3D finite-element electromagnetic simulation model of the 4.0 GHz radiating-edge gap-coupled patch antenna in Ansys HFSS.

III. Manufacturing & Experimental Setup

A. CNC Isolation Milling Prototyping Pipeline

Fabrication was executed using a low-cost, high-precision prototyping workflow:

1) **CAD/CAM Toolpathing:** Gerber vector files generated in KiCAD were processed in FlatCAM to compute isolation routing paths for a 0.2 mm 30° V-bit router.
2) **CNC Milling:** A 3-axis CNC router engraved isolation channels into single-sided FR-4 PCB stock ($35\,\mu$m copper cladding).

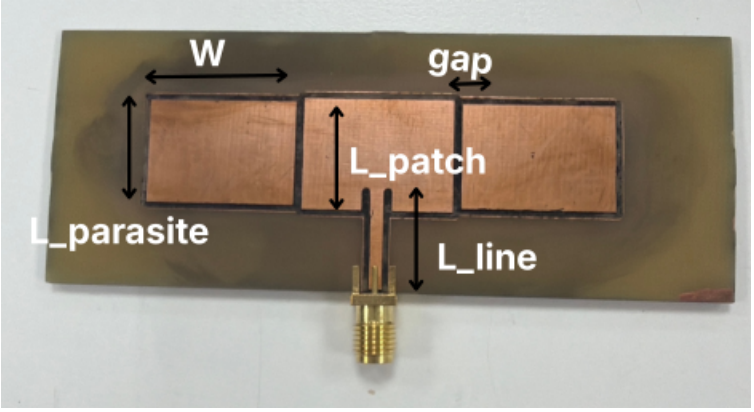


Fig. 2. Annotated physical prototype defining the driven patch length L_{patch}, parasitic patch length L_{parasite}, shared width W, coupling gap g, and feed line length L_{line}.

3) **Chemical Etch Finishing:** To remove thin residual copper bridges between the closely spaced gap ($g = 0.5$ mm) without damaging the trace boundaries, a quick immersion in ferric chloride ($FeCl_3$) was performed.
4) **SMA Connectorization:** An edge-mount $50\,\Omega$ SMA launcher was hand-soldered to the microstrip feed line and ground plane.

During CNC machining, two physical phenomena were identified as critical sources of structural error: end-mill kerf expansion due to runout and substrate trenching, where the router bit penetrates 0.1–0.2 mm into the FR-4 dielectric along trace borders.

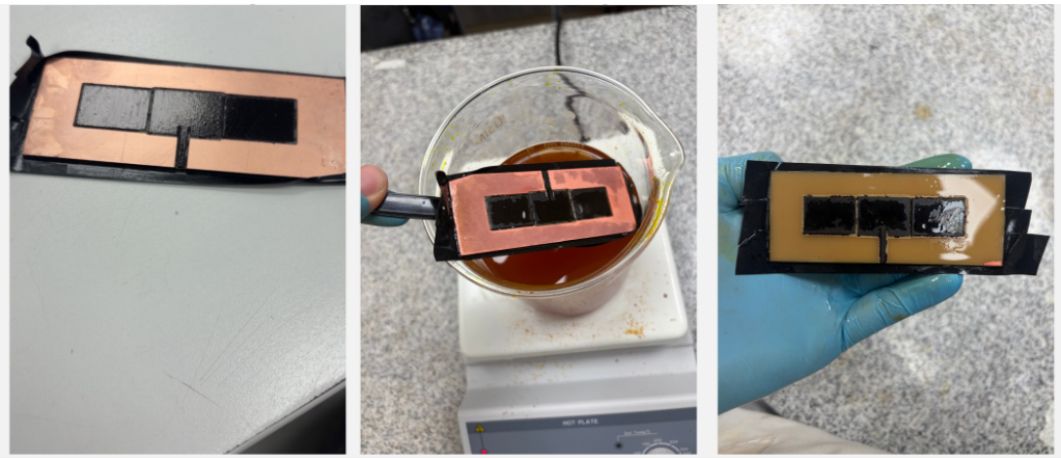


Fig. 3. CNC prototyping sequence showing isolation milling and masking (left), chemical etching immersion in ferric chloride ($FeCl_3$) (center), and final cleaned PCB prototype (right).

B. VNA Calibration & Custom 2D Far-Field Radiation Bench

Impedance characterization (S_{11}) and forward transmission (S_{21}) were conducted using a Rigol Vector Network Analyzer (VNA). Full vector error correction was established via standard Short-Open-Load-Thru (SOLT) calibration.

To extract far-field radiation patterns without requiring an expensive anechoic chamber or automated motorized rotators, a low-cost, hands-on DIY bench setup was engineered:

- **Fixed Transmitting (TX) Antenna:** Prototype 1 was repurposed as the fixed reference transmitting radiator, mounted on a low-permittivity styrofoam pedestal and connected to Port 1 of the Rigol VNA.
- **Receiving (RX) Antenna Under Test:** Prototype 2 (the optimized antenna element) served as the receiving Device Under Test (DUT), mounted on a mobile styrofoam pedestal and connected to Port 2 of the VNA.
- **Hand-Drawn 360° Protractor Grid:** An angular protractor template was hand-drawn on paper and secured to the lab bench to maintain constant radial separation and precise angular orientation.
- **Manual Angular S_{21} Measurements:** The RX antenna block was manually rotated around the protractor template in discrete angular steps of 20° in the principal E-plane and 10° in the principal H-plane, directly recording the transmission coefficient $S_{21}(\theta)$ on the VNA display at each angle.



Fig. 4. Custom DIY 2D far-field radiation pattern testbench: fixed TX reference antenna (Prototype 1, left), hand-drawn protractor paper grid, and manually rotated RX receiving antenna under test (Prototype 2, right).

IV. Experimental Results & Comparative Analysis

A. *Prototype 1 Failure Analysis: Dimensional Over-Cut*

Initial inspection of Prototype 1 revealed noticeable discrepancies between nominal HFSS geometry and physical dimensions post-milling, as recorded in Table II.

TABLE II
Geometric Comparison Across Design and Fabricated Prototypes

Dimension (mm)	HFSS Design	Prototype 1	Prototype 2
Feed Line Length (L_{line})	14.80	14.50	14.70
Parasitic Length (L_{parasite})	16.40	15.40	16.40
Driven Patch Length (L_{patch})	17.50	16.50	17.40
Patch Width (W)	22.80	22.10	22.70
Coupling Gap (g)	0.50	≈ 0.80	0.50

In Prototype 1, improper CNC bit offset configuration resulted in an effective -1.0 mm contraction of both patch lengths ($L_{\text{patch}} = 16.5$ mm, $L_{\text{parasite}} = 15.4$ mm) and an expansion of gap g to ≈ 0.8 mm.

As predicted by Eq. (5), shortening physical length L forces an upward shift in resonant frequency ($f_r \propto 1/L$). Furthermore, expanding gap g from 0.5 mm to 0.8 mm severely degraded inter-element mutual capacitance ($C_g \propto 1/g$). As shown in Table III, Prototype 1 suffered massive detuning: Peak 1 shifted from 3.84 GHz to 4.34 GHz ($+13.02\%$), and Peak 2 shifted from 4.06 GHz to 4.53 GHz

(+11.58%). Return loss deteriorated to −12.1 dB and −8.64 dB, collapsing the −10 dB operational bandwidth from 410 MHz (10.25%) down to just 120 MHz (3.00%).



Fig. 5. Vector Network Analyzer (VNA) S_{11} measurement for Prototype 1 demonstrating severe frequency detuning (+13.0%) and return loss degradation due to CNC over-cut errors.

TABLE III
COMPARATIVE IMPEDANCE MATCHING (S_{11}) PERFORMANCE: HFSS SIMULATION VS. PROTOTYPE 1 VS. PROTOTYPE 2

Performance Metric	**HFSS Sim.**	**Proto 1**	**Proto 1 Error**	**Proto 2**	**Proto 2 Error**
Low Cutoff ($S_{11} \leq -10$ dB)	3.745 GHz	4.280 GHz	+0.535 GHz (+14.3%)	3.900 GHz	+0.155 GHz (+4.1%)
High Cutoff ($S_{11} \leq -10$ dB)	4.155 GHz	4.400 GHz	+0.245 GHz (+5.9%)	4.270 GHz	+0.115 GHz (+2.8%)
−10 dB Bandwidth	410 **MHz** (10.3%)	120 **MHz** (3.0%)	−290 **MHz** (−70.7%)	370 **MHz** (9.3%)	−40 **MHz** (−9.8%)
Peak 1 Resonance Freq.	3.840 GHz	4.340 GHz	+0.500 GHz (+13.0%)	3.960 GHz	+0.120 GHz (+3.1%)
Peak 1 S_{11} Magnitude	−45.60 dB	−12.10 dB	+33.50 dB	−32.10 dB	+13.50 dB
Peak 2 Resonance Freq.	4.060 GHz	4.530 GHz	+0.470 GHz (+11.6%)	4.220 GHz	+0.160 GHz (+3.9%)
Peak 2 S_{11} Magnitude	−22.20 dB	−8.64 dB	+13.56 dB	−15.60 dB	+6.60 dB

B. Prototype 2 Validation & Return Loss Characterization

Following recalibration of toolpath isolation vectors and bit diameter compensation in FlatCAM, Prototype 2 achieved sub-tenth-millimeter accuracy ($L_{\text{patch}} = 17.4$ mm, $L_{\text{parasite}} = 16.4$ mm, $g = 0.5$ mm).

As detailed in Table III, Prototype 2 successfully restored wideband dual-resonance operation:

- Measured Peak 1 occurred at 3.96 GHz with $S_{11} = -32.1$ dB (+3.13% frequency shift from simulation).
- Measured Peak 2 occurred at 4.22 GHz with $S_{11} = -15.6$ dB (+3.94% frequency shift from simulation).
- The achieved −10 dB impedance bandwidth reached 370 **MHz** (9.25%), spanning 3.90 GHz to 4.27 GHz. This almost doubles the target specification of 200 MHz (5.0%) established for OFDM data transfer standards, providing substantial operational margin against environmental detuning and fabrication variance.

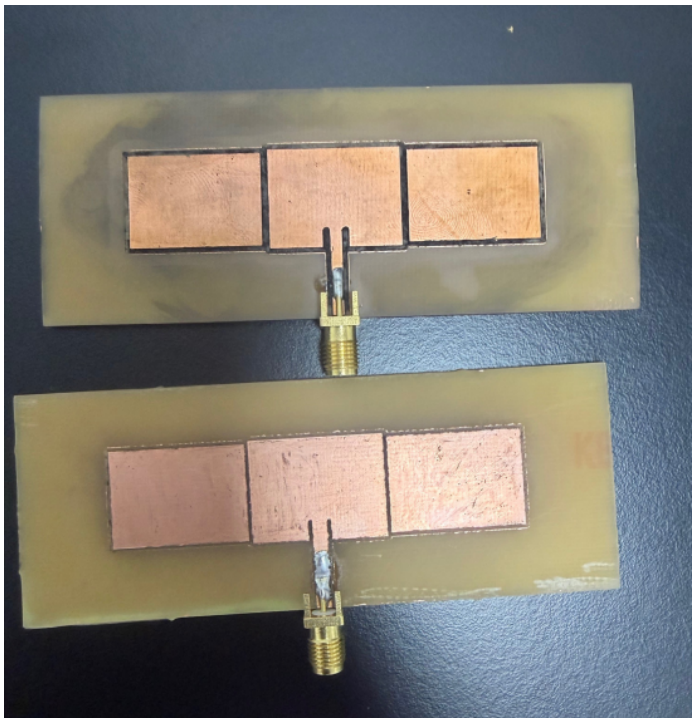


Fig. 6. Physical prototype comparison showing Prototype 1 (top) fabricated with sub-millimeter milling over-cuts and Prototype 2 (bottom) fabricated with recalibrated toolpaths.



Fig. 7. Vector Network Analyzer (VNA) S_{11} measurement for Prototype 2 confirming wideband dual-resonance response (370 MHz bandwidth, Peak 1 at -32.1 dB, Peak 2 at -15.6 dB).

The minor $+3$–4% systematic frequency shift observed between HFSS simulations and Prototype 2 is physically justified by two mechanisms:

1) **Dielectric Permittivity Dispersion:** Standard commercial FR-4 substrates exhibit batch-to-batch permittivity variations ($\epsilon_r \sim 4.15$–4.40 at 4 GHz). An effective permittivity of $\epsilon_r \approx 4.20$ in the manufactured board accounts directly for a $+3.2\%$ frequency elevation according to $f_r \propto 1/\sqrt{\epsilon_r}$.
2) **Substrate Engraving/Trenching:** Isolation milling removes dielectric material immediately bordering the copper edges. Replacing FR-4 ($\epsilon_r = 4.4$) with air ($\epsilon_r = 1.0$) in the fringing field channels lowers ϵ_{eff}, further elevating resonant frequencies.

C. Far-Field Radiation Patterns & Beamwidth Analysis

Measured normalized radiation patterns for Prototype 2 in the principal H-plane and E-plane are presented in Table IV.

TABLE IV
MEASURED FAR-FIELD RADIATION PERFORMANCE OF PROTOTYPE 2

Resonance Peak	**Plane Cut**	**HPBW (°)**	**FBR (dB)**
Peak 1 (3.96 GHz)	H-Plane	49.0°	14.0 dB
Peak 1 (3.96 GHz)	E-Plane	96.0°	15.7 dB
Peak 2 (4.22 GHz)	H-Plane	36.0°	14.0 dB
Peak 2 (4.22 GHz)	E-Plane	95.0°	14.0 dB

The antenna displays directive broadside beam patterns across the entire operational band. In the E-plane, half-power beamwidths ($HPBW$) remain stable at $96.0°$ (Peak 1) and $95.0°$ (Peak 2). In the H-plane, the beamwidth narrows from $49.0°$ at 3.96 GHz to $36.0°$ at 4.22 GHz due to increased electrical aperture width (W/λ_0) at higher frequencies. Front-to-Back Ratios (FBR) exceed 14.0 dB across both planes.

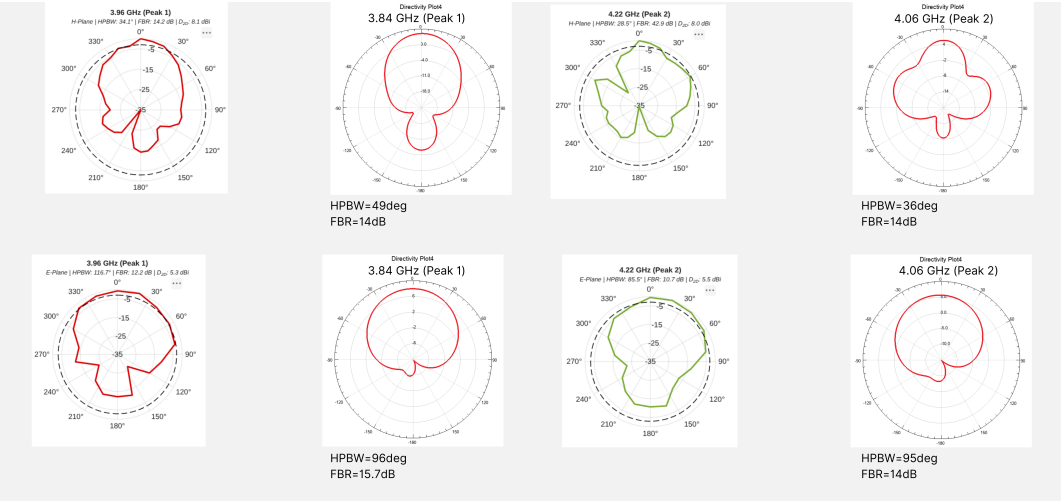


Fig. 8. Measured 2D far-field polar radiation pattern cuts for Prototype 2: H-plane at Peak 1 (top-left) and Peak 2 (top-right); E-plane at Peak 1 (bottom-left) and Peak 2 (bottom-right).

D. Directivity Discrepancy Discussion

A critical analytical observation during testing was the apparent contradiction between measured 2D directivity ($D_{2D} \approx 8.0$ dBi) and simulated 3D full-sphere directivity ($D_{3D} = 5.65$–7.20 dBi). This discrepancy is rigorously explained through mathematical and physical bench modeling:

1) Mathematical Single-Cut Integration Error: Standard 2D directivity estimation formulas (such as the Kraus approximation) compute directivity exclusively from principal plane half-power beamwidths (Θ_E and Θ_H in degrees):

$$D_{\text{Kraus}} = \frac{41253}{\Theta_E^\circ \cdot \Theta_H^\circ} \tag{6}$$

Evaluating Eq. (6) with measured beamwidths yields:

- At 3.96 GHz ($\Theta_E = 96^\circ, \Theta_H = 49^\circ$):

$$D_{\text{Kraus}} \approx \frac{41253}{96 \times 49} = 8.76 \implies 9.42 \text{ dBi}$$

- At 4.22 GHz ($\Theta_E = 95^\circ, \Theta_H = 36^\circ$):

$$D_{\text{Kraus}} \approx \frac{41253}{95 \times 36} = 12.06 \implies 10.81 \text{ dBi}$$

In contrast, exact 3D directivity requires integrating radiation intensity $U(\theta, \phi)$ across the complete 4π sr solid angle:

$$D_{3D} = \frac{4\pi U_{\max}}{P_{\text{rad}}} = \frac{4\pi U_{\max}}{\int_0^{2\pi} \int_0^{\pi} U(\theta, \phi) \sin\theta \, d\theta d\phi} \tag{7}$$

Approximating 3D solid angle power integration via orthogonal 2D cuts assumes azimuthal pattern symmetry and completely neglects off-axis nulls, minor side-lobes, and cross-polarized field components. Consequently, single-cut integration undercounts back-hemisphere power, artificially inflating computed front-side directivity by 1.5–2.5 dB.

2) Testbench Mount & Coaxial Shielding Artifacts: In the physical setup, the prototype antenna was mounted on a metallic support arm with a rigid coaxial cable extending directly behind the ground plane. This metallic structure acts as a secondary reflector, suppressing backward radiation ($FBR \approx 14$–16 dB). By artificially attenuating back-lobe power $\iint_{\text{back}} U d\Omega$, the physical bench structure forces a larger fraction of total radiated power into the forward hemisphere, raising the measured forward directivity relative to free-space HFSS numerical models.

V. Lessons Learned & Engineering Takeaways

Key engineering takeaways established during this design and fabrication cycle include:

1) **Tight Sensitivity of High-Q Coupling Gaps:** Parasitic gap-coupled resonators are extremely sensitive to sub-millimeter manufacturing tolerances (± 0.1 mm). Isolation toolpath kerf width must be calibrated prior to final milling.
2) **Dielectric Trenching Compensation:** Substrate removal around trace borders elevates resonant frequencies. Numerical models on low-cost FR-4 should include 0.1 mm dielectric trenching or adjust ϵ_r downward by 3–4%.
3) **Validation of 2D Radiation Bench Limits:** Single-cut 2D polar measurements provide robust relative beamwidth and null locations, but over-estimate absolute directivity due to coordinate truncation and setup reflections.

VI. Future Work & Perspectives

To prepare this antenna technology for deployment in USFQ's 5G ToA positioning and MIMO RF fingerprinting testbed, future work will focus on:

- **Substrate Upgrade:** Transitioning from high-loss FR-4 ($\tan\delta = 0.02$, efficiency $\approx 63\%$) to low-loss RF laminates such as Rogers RO4003C ($\epsilon_r = 3.55, \tan\delta = 0.0027$), boosting radiation efficiency beyond 90%.
- **Corporate-Fed Array Design:** Scaling the validated gap-coupled element into 2×2 and 4×4 corporate-fed microstrip patch arrays to achieve narrow main beams ($HPBW < 20^\circ$) and realized gain exceeding 12 dBi.
- **System Integration:** Interfacing custom antenna arrays with USRP B210 SDR platforms to execute real-time angle-of-arrival (AoA) estimation and active beam-steering algorithms.

VII. Conclusion

This report successfully demonstrated the end-to-end design, simulation, prototyping, and experimental characterization of a wideband 4.0 GHz gap-coupled microstrip patch antenna. By pairing an inset-fed driven patch with a coplanar parasitic resonator, an impedance bandwidth of 370 MHz (9.25%) was experimentally achieved on low-cost FR-4 substrate, nearly doubling the 200 MHz (5.0%) bandwidth requirement specified for standard OFDM data transfer. Root-cause failure analysis of initial fabrication errors enabled precise CNC toolpath compensation in Prototype 2, yielding excellent agreement between HFSS predictions and VNA measurements. Detailed radiation pattern characterization confirmed directive broadside performance and provided rigorous physical justifications for 2D vs. 3D directivity discrepancies. The engineered antenna provides a robust front-end element for advanced 5G beamforming and RF positioning research.

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